INCH-POUND
MIL-DTL-24643/11E
1 October 2009
SUPERSEDING
MIL-DTL-24643/11D
22 August 2002

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -20 °C TO +90 °C, 600 VOLTS, TYPE LSSSF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction, Non-Watertight

First	-	Copper conductor, uncoated, 300 MCM AWG, Class G stranding (see note 1).
Second	-	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	-	Thermoset insulation. Minimum average wall thickness of 0.075 inch.
Fourth	-	Optional reinforcement.
Fifth	-	Cross-linked polyolefin jacket bonded to the underlying insulation. Minimum average wall thickness of 0.075 inch.

NOTE: The copper shall be hard drawn, having a minimum tensile strength of 67,200 lb/in, and minimum elongation of 0.85 percent in 60-inch lengths before stranding. The resistance specified is based on 102.7 percent of the values required for annealed bare copper.

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TABLE I. Details.

Military part no. M24643/11	Type and size	Conductor size	Overall diameter	
			min. (inch)	max. (inch)
-01UN	LSSSF-7	300 MCM (Class G)	1.020	1.100

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic Electricals:	
Conductor resistance (ohms/1000 feet at 25 °C, max.)	0.0388
Voltage withstand (volts, root mean square, min.)	
Conductor to ground	2500
Insulation resistance (megohms/1000 feet, min.)	
Conductor to ground	100
Conductor continuity	No failure
Group A:	
Visual and dimensional	No failure
Group B:	
Cold bending cable at minus 30±2 °C two turns around an 8.5" mandrel	No damage
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 200 °C)	50
Drip (95±1 °C)	Zero
Physicals (unaged)	
Insulation (extruded)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35

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Group C:	
Physicals (aged) air oven	
Insulation (extruded)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Shrinkage	No failure
Group D:	
Flame propagation (cable)	No failure
Qualification Inspection:	
Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:	
Aging and compatibility (cable) (125±5 °C)	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Insulation	45
Toxicity index (max.)	
Jacket	5
Insulation	1.5
Durometer (jacket) - Type A (hardness, min.)	80

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Weathering (jacket)	No failure
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (minimum)	Feet (maximum)
LSSSF-7	1045	1155

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2008-015)

Review Activities:
Army – AR, AV, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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